

Linear Regression Interview Questions & Answers

Basic Level

1. What is Linear Regression?

Answer:

Linear Regression is a statistical method used to model the relationship between a dependent variable and one or more independent variables by fitting a linear equation to observed data.

2. What are the types of Linear Regression?

Answer:

- Simple Linear Regression (one independent variable)
- Multiple Linear Regression (two or more independent variables)

3. What is the equation of a simple linear regression?

Answer:

$$y = 0 + 1x +$$

Where:

- y: dependent variable
- x: independent variable
- 0: intercept
- 1: slope
- : error term

4. How do you interpret the coefficients in linear regression?

Answer:

- Intercept (0): Value of y when x = 0.
- Slope (1): Change in y for a one-unit change in x.

5. What is the purpose of the cost function in Linear Regression?

Answer:

To measure the difference between predicted and actual values. The most common cost function is Mean Squared Error (MSE).

Intermediate Level

6. What are assumptions of Linear Regression?

Answer:

1. Linearity
2. Independence
3. Homoscedasticity (constant variance of error terms)
4. Normal distribution of error terms
5. No multicollinearity (for multiple regression)

7. How is linear regression optimized?

Answer:

By minimizing the cost function using Ordinary Least Squares (OLS) or Gradient Descent.

8. What is R-squared (R)?

Answer:

A statistical measure that indicates how well data fit a regression model.

$$R = 1 - (SS_{\text{res}} / SS_{\text{tot}})$$

Ranges from 0 to 1.

9. What are some evaluation metrics for linear regression?

Answer:

- Mean Absolute Error (MAE)
- Mean Squared Error (MSE)
- Root Mean Squared Error (RMSE)
- R-squared (R)

10. What is multicollinearity and how do you detect it?

Answer:

Multicollinearity occurs when independent variables are highly correlated.

Detection: Use Variance Inflation Factor (VIF). If $VIF > 5$ or 10 , multicollinearity may be a problem.

Advanced Level

11. What is the difference between L1 and L2 regularization?

Answer:

- L1 (Lasso): Adds absolute value of coefficients to the cost function; can shrink some coefficients to 0.
- L2 (Ridge): Adds squared value of coefficients to the cost function; shrinks coefficients but doesn't eliminate them.

12. What is adjusted R-squared?

Answer:

Adjusted R adjusts the R value based on the number of predictors in the model.

$$\text{Adjusted R} = 1 - [(1 - R)(n - 1)/(n - k - 1)]$$

13. What is heteroscedasticity? How do you detect it?

Answer:

Heteroscedasticity is when the variance of residuals is not constant.

Detection: Residual plots or statistical tests like Breusch-Pagan.

14. What is the difference between Linear Regression and Logistic Regression?

Answer:

- Linear Regression: Predicts continuous values.
- Logistic Regression: Predicts probabilities or binary outcomes.

15. What if the relationship is not linear?

Answer:

You can:

- Transform variables (e.g., log, sqrt)
- Use Polynomial Regression
- Try other models like decision trees or SVM

16. Explain the role of the error term in Linear Regression.

Answer:

The error term accounts for the variability in the response variable that cannot be explained by the model.

17. When would you prefer Ridge over Lasso Regression?

Answer:

- Ridge: When all features are potentially useful and multicollinearity is present.
- Lasso: When feature selection is desired.

18. Can Linear Regression be used for time series data?

Answer:

Yes, but only with careful preprocessing. Autocorrelation and non-stationarity need to be addressed.

19. What are interaction terms in Linear Regression?

Answer:

Variables created by multiplying two or more predictors to capture combined effects.

20. How do you handle outliers in Linear Regression?

Answer:

- Remove or transform them
- Use Robust Regression
- Check influence using Cooks Distance